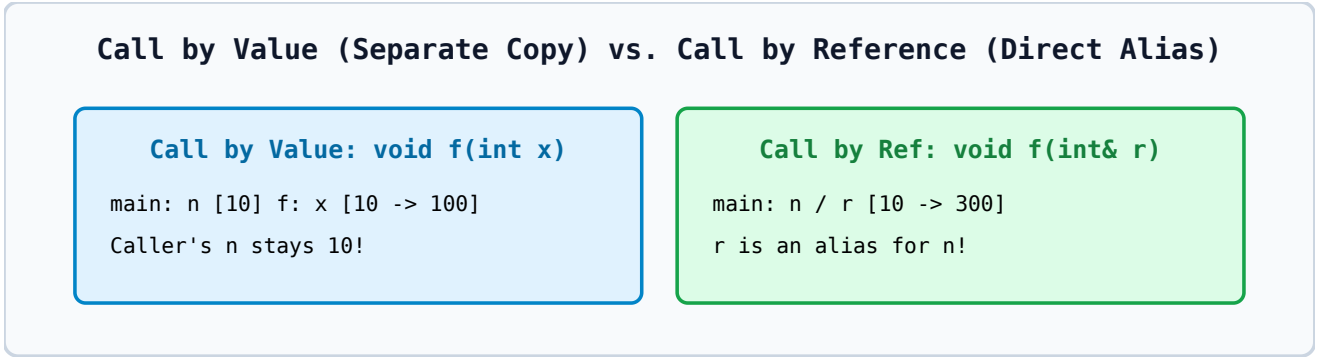
05 FUNCTIONS • THEORY

01 parameter passing value address reference

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

01 — Call by Value, Call by Address (Pointer), and Call by Reference



Call by Value vs Reference

Mechanism	Syntax	Copies Argument?	Can Modify Caller's Variable?
Call by Value	`void f(int x)`	**Yes** (independent local copy)	**No**
Call by Address	`void f(int* p)`	Copies pointer address (8 bytes)	**Yes** (via `*p = ...`)
Call by Reference	`void f(int& r)`	**No** (`r` aliases caller's variable)	**Yes** (via `r = ...`)
Call by `const` Reference	`void f(const string& s)`	**No**	**No** (Read-only, zero-copy)